

Scalable Biomechanical Modeling Through Federated Learning and Resource Federation

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Quantitative Evaluation & Design (QED) Group



General Application Domains

- ## Critical Example: Biomechanical Signals

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Dangi et al. (2025), Efficient DL for Muscle Activity Patterns.

Estimating Biomechanical Signals Using Simple Models

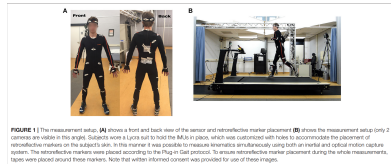
Ground Truth (lab-based, controlled environments)

- GRFs measured by force plates / instrumented treadmills
- Motion capture provides segment kinematics
- Joint moments via inverse dynamics: GRFs + kinematics

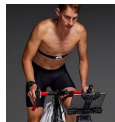
Estimation from Wearables

- IMUs: linear acceleration and angular velocity
- EMG records muscle activation
- GRFs/joint moments estimated indirectly via *physics-based* models or *data-driven* approaches

Much less expensive, freedom of movement, works in the wild.



F. J. Wouda et al., *Front. Physiol.*, 2018.



Overview

Part I (Modeling) — Estimating Ground Reaction Forces From Inertial Sensors

Edge deployment constraint: must operate on edge devices; deep networks are often too costly.



Part II (Data) — Federated Multi-Task Learning for Estimating Biomechanical Signals from Wearable Sensors

Data distribution & privacy constraint: data are distributed and heterogeneous; pooling for centralized training breaks privacy.



Part III (Resource) — Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements

Training computational-resource constraint: uneven and persistent computation needs; need guarantees of mutual benefit.



Future Work

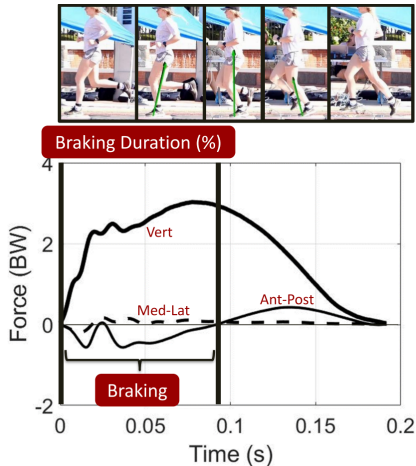
Part I: Estimating Ground Reaction Forces From Inertial Sensors

Song, B., Paolieri, M., Stewart, H. E., Golubchik, L., McNitt-Gray, J. L., Misra, V., & Shah, D. (2025).
Estimating Ground Reaction Forces From Inertial Sensors. IEEE Trans. Biomed. Eng., 72(2).

Motivation

- **Goal:** predict signals of interest from cheap-to-collect signals.
- ML methods can
 - facilitate such data collection “in the wild”
 - monitor for signals that suggest injury risk, analyze athletic performance, etc.

We want simple, cheap, explainable ML techniques.



Feat. Harper E. Stewart for graph.

High-Dimensional Signals from Multi-Modal Sensors

$$A \in \mathbb{R}^{n \times T \times L \times M \times D}$$

- n : Segments T : Time L : Locations M : Modalities D : Directions

Problem: learn a mapping $f : A_{\text{in}} \rightarrow A_{\text{out}}$ to predict time-series trajectories from complex sensor data.

- **Input** (A_{in}): high-dimensional, multi-modal sensor grid defined by $\{T_{\text{in}}, L_{\text{in}}, M_{\text{in}}, D_{\text{in}}\}$.
- **Output** (A_{out}): targeted biomechanical time-series defined by $\{T_{\text{out}}, L_{\text{out}}, M_{\text{out}}, D_{\text{out}}\}$.

Challenges

- **Degraded signal quality:** noise, sparse data with missing entries.

Field specific

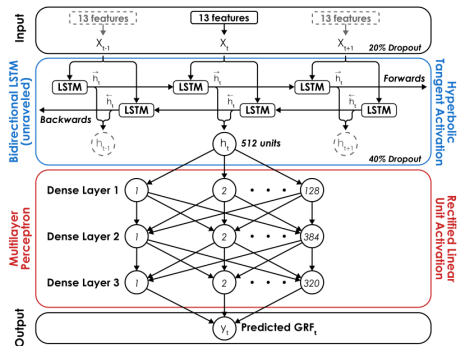
- **Limited ground truth:** expensive, instrumented data collection.

Problem specific

- **Temporal dependence:** current ground contact depends on previous ones.
- **Multi-modal sensing:** redundancy, complementarity, and interference.
- **Phase-dependent relevance:** sensors dominate at different movement phases.

State of the Art: Deep Neural Network

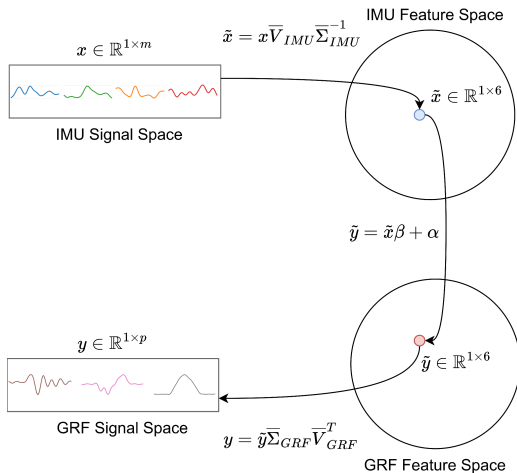
State-of-the-art neural networks $\Rightarrow$ expensive to train, lacking interpretability.



R. S. Alcantara et al., "Predicting continuous ground reaction forces from accelerometers during uphill and downhill running: A recurrent neural network solution," *PeerJ*, 2022.

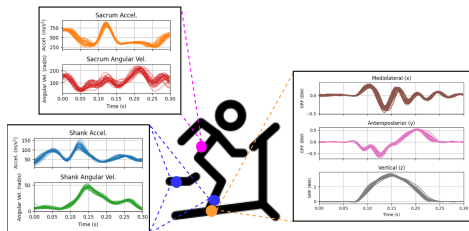
Key Idea — SVD Embedding Regression (SER)

- **Dimensionality reduction:** compress both input (IMU) and output (GRF) signals into low-dimensional embeddings.
- **Learn mapping in embedding space:** a simple regression between the low-dimensional representations.
- **Predict in low dimension:**
 - project inputs to lower embeddings
 - predict output embeddings
 - reconstruct the full signal



Pac-12 Dataset

- 44 competitive collegiate distance runners from 3 universities (25 female, 19 male).
- 114 instrumented-treadmill collections (1–6 per athlete), each with 100–200 trials.
 - Male: 7, 6.5, 5 min/mi (3.8–5.4 m/s)
 - Female: 7, 5.5 min/mi (3.8–4.9 m/s)
- **Sensors**
 - Instrumented treadmills: 3D Ground Reaction Forces (GRF)
 - Wearable IMUs: acceleration + angular velocity at sacrum, left shank, right shank
- Signals aligned and split into 400 ms windows (200 time points), one foot contact per window.



Data collected by biomechanics domain experts, including Prof. Jill McNitt-Gray and Dr. Harper E. Stewart.

Making a Low-Rank Signal Matrix Arrangement

Multi-Modal Motion Segments

$$A \in \mathbb{R}^{n \times T \times L \times M \times D}$$

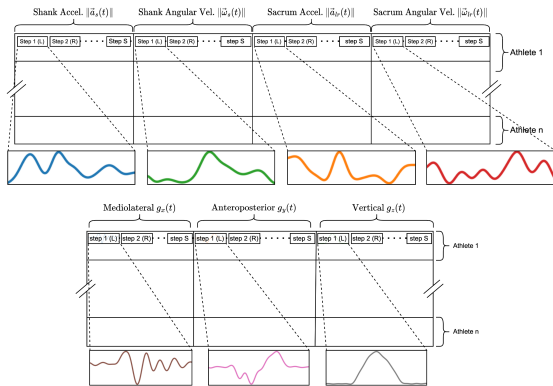
- T : time, L : sensor locations
- M : modalities, D : directions

Full Concatenation. Each row is a complete motion context:

$$x_i = \text{vec}(A_i) \in \mathbb{R}^{TLMD} \quad (\text{Segment})$$

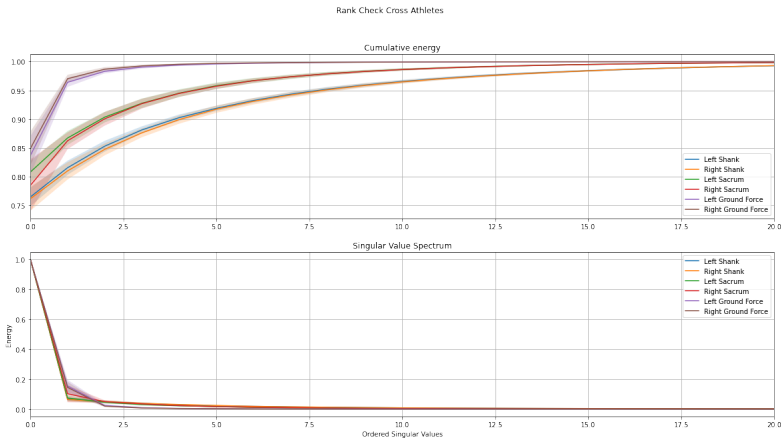
Stacking (tunable). Concatenate $S \geq 1$ consecutive segments:

$$X = \begin{bmatrix} x_1^T & \cdots & x_S^T \\ x_2^T & \cdots & x_{S+1}^T \\ \vdots & & \vdots \end{bmatrix} \in \mathbb{R}^{(n-S+1) \times (STLMD)}$$



Low-Rank Data Structure

- Input and output ranks are hyperparameters.
- Chosen to represent at least **95%** of the matrix energy.



Low-Rank Representation and Reconstruction

Singular Value Decomposition of a matrix $A \in \mathbb{R}^{n \times d}$:

$$A = U \Sigma V^\top$$

Optimal low-rank approximation:

$$\min_{\text{rank}(B) \leq r} \|A - B\|_F^2 = \|A - U^{(r)} \Sigma^{(r)} V^{(r)\top}\|_F^2$$

Low-rank embedding (projection):

$$U^{(r)} = A V^{(r)} (\Sigma^{(r)})^{-1}$$

Reconstruction:

$$A \approx U^{(r)} \Sigma^{(r)} V^{(r)\top}$$

SVD Embedding Regression (SER)

Embedding (coordinate) representation

$$\tilde{x}_i = x_i V_X^{(r)} (\Sigma_X^{(r)})^{-1} \in \mathbb{R}^r, \quad \tilde{y}_i = y_i V_Y^{(r)} (\Sigma_Y^{(r)})^{-1} \in \mathbb{R}^r$$

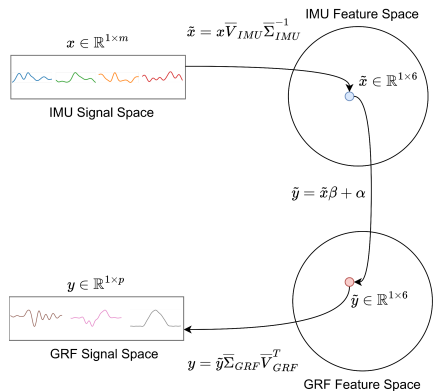
Subspace-constrained operator learning

$$W = \arg \min_W \sum_{i=1}^n \|\tilde{y}_i - W \tilde{x}_i\|_2^2 + \lambda_2 \|W\|_F^2 + \lambda_1 \|W\|_1$$

Inference with physically valid reconstruction

$$\hat{y} = (\tilde{x} W + b) \Sigma_Y^{(r)} V_Y^{(r)\top}$$

- Learns a low-rank mapping *between signal subspaces*.
- Regression constrained by learned IMU and GRF bases.
- Reconstruction enforces structured, physically plausible outputs.



Accuracy of Lightweight Models

RMSE (IN BODY WEIGHT UNITS, BW, I.E., N/KG) OF VERTICAL GRF ESTIMATIONS $RMSE(g_z, \hat{g}_z)$ FOR DIFFERENT INPUT SIGNALS, DATA SCENARIOS, MACHINE LEARNING METHODS

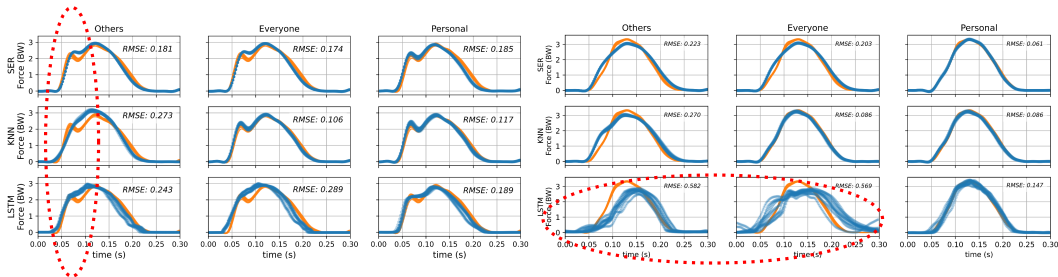
Input Signals	Scenario OTHERS			Scenario EVERYONE			Scenario PERSONAL		
	SER	KNN	LSTM	SER	KNN	LSTM (FINE-TUNED)	SER	KNN	LSTM
ALL	0.197	0.180	0.126	0.187	0.118	0.124 (0.117)	0.130	0.122	0.134
ACC	0.220	0.197	0.177	0.210	0.125	0.175 (0.151)	0.127	0.127	0.143
ANG	0.197	0.187	0.183	0.190	0.130	0.183 (0.180)	0.133	0.132	0.171
SHANK	0.215	0.210	0.206	0.205	0.149	0.209 (0.190)	0.139	0.134	0.188
SACRUM	0.217	0.210	0.289	0.205	0.129	0.286 (0.270)	0.136	0.137	0.184
SAC/ACC3D	0.194	0.181	0.171	0.185	0.122	0.177 (0.160)	0.128	0.132	0.178
SAC/ACC	0.198	0.190	0.187	0.190	0.130	0.206 (0.177)	0.129	0.133	0.193

Results highlighted in blue are optimal or less than 0.010 from optimal for a scenario and set of input signals.

Baseline LSTM after R. S. Alcantara et al., *PeerJ*, 2022. Best result per data regime highlighted.

Physical Plausibility and Clinical Relevance in GRF Estimation

- Clinically important GRF features — **impact peak** (missing or falsely present) and **rate of force rise** — are critical for gait analysis.
- Physically meaningful structure matters: LSTM predictions may produce **non-physical artifacts** (unrealistic oscillations or distorted waveforms).



Deployment on GPU and Mobile Device (Inference)

Samsung Galaxy S20 (Exynos, 2.73 GHz) · 10-step GRF inference

	LSTM		KNN	SER
Latency	14.9 s		36 ms	0.3 ms
Memory	180 MB	170 MB (global) / 9 MB (personal)		4 MB

SER runs in 0.3 ms within 4 MB — real-time, on-device inference; LSTM's ~15 s makes real-time estimation impractical.

Compare to State of the Art

COMPARISON WITH STATE-OF-THE-ART STUDIES ON GRF ESTIMATION. RESULTS ARE REPORTED FOR EACH OF OUR METHODS (SER, KNN, LSTM) WHEN USING ALL INPUT SIGNALS IN THEIR BEST SCENARIOS OF APPLICATION (PERSONAL, EVERYONE, OTHERS, RESPECTIVELY); OTHER STUDIES CONSIDER THE OTHERS SCENARIO

Paper	GRF Measurement	Sensors Type	Sensor Locations	Estimation Model	RMSE (BW)			rRMSE (%)		
					<i>x</i>	<i>y</i>	<i>z</i>	<i>x</i>	<i>y</i>	<i>z</i>
[22]	Instrumented treadmill, multiple speeds and slopes	IMU (acceleration)	Sacrum	LSTM	—	—	0.16	—	—	6.4
[37]	In-sole sensors, multiple speeds	IMU (acceleration)	Left/right shanks	MLP	—	—	0.15	—	—	—
[19]	Force plate, multiple speeds	IMU (acceleration, angular velocity)	Sacrum, right thigh, shank, foot	Simulation + CNN	—	—	—	—	2	6
[20]	Instrumented force plates and treadmills	Video, IMU (acceleration)	Sacrum, left/right shanks, thighs	CNN	—	—	—	21.6	17.1	13.9
[38]	Instrumented treadmill, multiple speeds	Video, IMU (acceleration)	Sacrum, left/right shanks	MLP	—	—	0.27	—	—	—
[39]	Instrumented treadmill, heel strike runners	IMU (acceleration)	Feet, (proximal) tibias, thighs, pelvis, and trunk	Physical model	0.05	0.07	0.18	10.8	7.8	6.6
Ours	Instrumented treadmill, multiple speeds	IMU (acceleration, angular velocity)	Sacrum, left/right shanks	SER (PERSONAL)	—	0.05	0.13	—	5.0	4.2
				KNN (EVERYONE)	—	0.05	0.12	—	4.6	3.9
				LSTM (OTHERS)	—	0.06	0.13	—	7.1	4.2

Summary of Contributions

- Proposed a novel **SVD Embedding Regression (SER)** framework for estimating GRFs from wearable IMUs collected at the shank and sacrum.
- Lightweight ML methods (SER, KNN) achieve accuracy comparable to or exceeding deep networks such as LSTMs, while being substantially more efficient.
- Jointly leveraging acceleration and angular velocity across multiple sensor locations yields the best GRF estimation performance.
- Incorporating personal training data significantly reduces estimation error across methods.

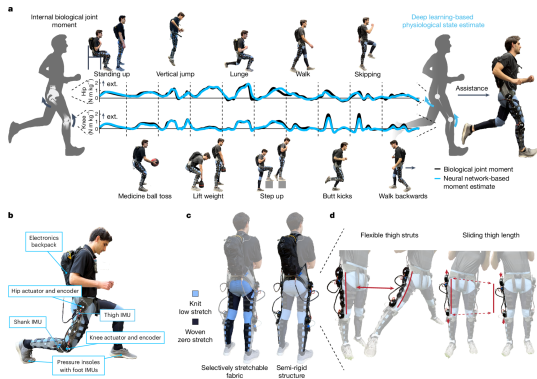
Part II: Federated Multi-Task Learning for Biomechanical Signals

Multi-Task Nature of Human Movement

- Human movement consists of many distinct activities, not a single task.
- Activities differ substantially in dynamics, joint loading, etc.:
 - walking, running, stairs, sit-to-stand, squatting, jumping, cutting

Key Challenge

- In real-world use, unseen or rare activities are inevitable.



Molinario, D. D., et al. "Task-agnostic exoskeleton control via biological joint moment estimation." *Nature* 635.8038 (2024): 337–344.

The Need for Going Beyond Plain SER

- For **unseen actions**, pooling data across all subjects (“Everyone”) generally improves performance — motivating federated learning for privacy (see 1st row of results).
- **One model per action** consistently outperforms a single global model across all actions (3rd row RMSE consistently better than 2nd row).

Table 1: SER Performance (RMSE) for Hip and Knee Moment estimation under different evaluation protocols. (results highlighted in blue are optimal or less than 0.010 from optimal for a scenario)

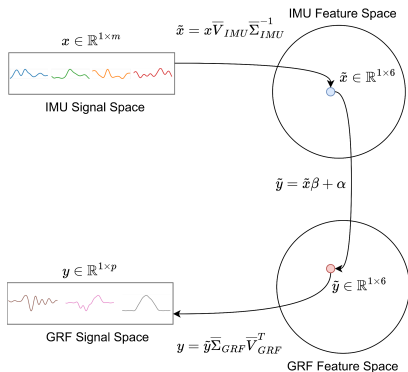
Evaluation Scenarios	Hip Flexion Moment			Knee Angle Moment		
	Personal	Everyone	Others	Personal	Everyone	Others
Leave-One Action-Out	0.235	0.228	0.230	0.218	0.186	0.188
One Model All Actions	0.146	0.181	0.190	0.118	0.147	0.155
One Model Per Action	0.103	0.111	0.159	0.089	0.094	0.135

Challenges

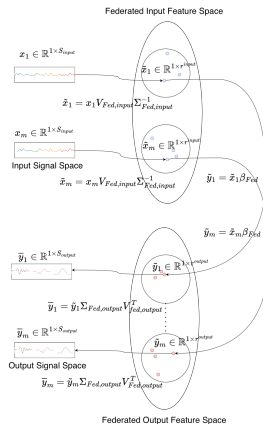
- **Need Federated SER**
 - preserve client privacy while learning from all data
- **Need SER to be one-model-per-action**
 - labels of what action it is are not always available

From SER to Federated SER

Both SER components — the SVD and the linear regression — must be learned in a federated manner, without sharing raw data.



Centralized SER



Federated SER

Federated SER Problem Definitions: SVD

Federated SVD

Let there be C clients. Client $i \in \{1, \dots, C\}$ holds a private data matrix

$$M = \begin{bmatrix} M_1 \\ \vdots \\ M_C \end{bmatrix}, \quad M_i \in \mathbb{R}^{n_i \times d}.$$

The goal is to compute a rank- k global factorization

$$M_i \approx U_i \Sigma_{\text{Fed}} V_{\text{Fed}}^T,$$

$$V_{\text{Fed}} \in \mathbb{R}^{d \times k}, \quad \Sigma_{\text{Fed}} \in \mathbb{R}^{k \times k},$$

without sharing raw data across clients.

Federated Linear Regression

Let there be C clients. Client $i \in \{1, \dots, C\}$ holds

$$\tilde{X}_i \in \mathbb{R}^{n_i \times r_{\text{input}}}, \quad \tilde{Y}_i \in \mathbb{R}^{n_i \times r_{\text{output}}}.$$

The goal is to learn a global regression weight $\beta_{\text{Fed}} \in \mathbb{R}^{r_{\text{input}} \times r_{\text{output}}}$ that minimizes the aggregated loss

$$\sum_{i=1}^C \|\tilde{X}_i \beta_{\text{Fed}} - \tilde{Y}_i\|_F^2,$$

without sharing raw embeddings.

SVD via Gradient Update — Learning V

Rank-1 (single-component) approximation loss

$$\mathcal{L}(x; A) = \frac{1}{4} \|A - xx^\top\|_F^2, \quad x \in \mathbb{R}^d.$$

This objective extracts a single dominant component of A and serves as the atomic building block of block-based k -SVD. Assuming A is a **symmetric, square** matrix.

$$\text{Gradient: } \nabla \mathcal{L}(x) = \|x\|_2^2 x - Ax.$$

General gradient-descent formulation: $x_{t+1} = x_t - \alpha_t \nabla \mathcal{L}(x_t)$. Choosing $\alpha_t = \eta / \|x_t\|_2^2$ as an adaptive step size gives

$$x_{t+1} = (1 - \eta) x_t + \frac{\eta}{\|x_t\|_2^2} Ax_t.$$

Key property: the update depends only on matrix–vector products Ax_t .

Jedra, Y., and Shah, D. “k-SVD with Gradient Descent.” NeurIPS 2025 Workshop MLxOR.

Federated SVD via Gradient Update — Learning V_{Fed}

Let each client $i \in \{1, \dots, C\}$ hold a private data matrix $M_i \in \mathbb{R}^{m_i \times d}$, where

$M \triangleq [M_1^\top \ \dots \ M_C^\top]^\top$. Define the global normal matrix

$$A \triangleq M^\top M = \sum_{i=1}^C M_i^\top M_i \in \mathbb{R}^{d \times d}, \quad A \succeq 0,$$

which is never explicitly constructed.

Instead, A is accessed only through matrix–vector products:

$$AX \triangleq \sum_{i=1}^C M_i^\top (M_i X), \quad X \in \mathbb{R}^{d \times k}.$$

Why this yields the SVD: if $M = U\Sigma V^\top$, then

$$A \triangleq M^\top M = V\Sigma^2 V^\top.$$

Using Rayleigh quotients,

$$\sigma_j^2 = v_j^\top A v_j \Rightarrow \sigma_j = \sqrt{v_j^\top A v_j}.$$

Federated SVD — Block-Based Update

Block update:

$$X_{t+1} = (1 - \eta) X_t + \eta A X_t$$

Orthonormalization:

$$X_{t+1} \leftarrow \text{orth}(X_{t+1}), \quad X_{t+1}^\top X_{t+1} = I_b$$

- The block size b serves as an *acceleration parameter*, enabling joint refinement of a b -dimensional subspace.
- Updating multiple directions together amortizes normal-operator evaluations and reduces wall-clock convergence time.
- Deflation separates stabilized dominant directions while allowing remaining components to keep evolving jointly under different stopping criteria.

SVD: Standard Rank-1 Deflation

Standard SVD. Given a data matrix $A \in \mathbb{R}^{n \times d}$,

$$A = \sum_{i=1}^k \sigma_i u_i v_i^\top, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k.$$

Rank-1 structure. The dominant pattern in the data is captured by

$$A_1 = \sigma_1 u_1 v_1^\top.$$

Rank-1 deflation. To learn additional structure, remove the explained component:

$$A^{(1)} = A - \sigma_1 u_1 v_1^\top.$$

- Do this iteratively for each rank from 1 to k .
- Deflation ensures subsequent components are learned from the *remaining* data variance, not from directions already captured.

Jedra, Y., and Shah, D. “k-SVD with Gradient Descent.” NeurIPS 2025 Workshop MLxOR.

Federated Block-Based Deflation

Important distinction. We do *not* apply deflation to the raw client data. Instead, deflation is applied to the **current subspace estimate**.

Server-side. Let $X \in \mathbb{R}^{d \times b}$ denote the candidate block of new directions to be added to the subspace.

Block deflation. Given the previously learned (e.g. 2) block of subspace $V \in \mathbb{R}^{d \times 2b}$, $V^\top V = I$, we remove overlap with existing directions:

$$X_\perp = (I - VV^\top)X$$

Why this matters.

- Deflation is done on the **server side** (lessens operations on clients).
- Deflation is done *once per round* on the server, ensuring global consistency.

Federated SER Problem Definitions: Linear Regression

Federated Linear Regression

Let there be C clients. Client $i \in \{1, \dots, C\}$ holds samples from the low-dimensional embedding space

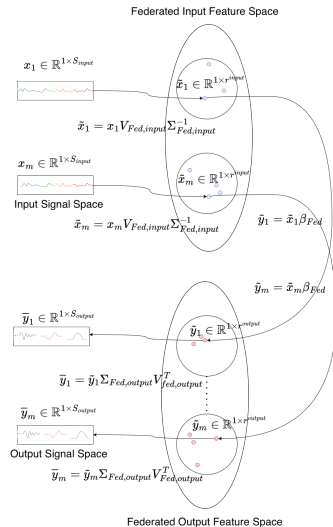
$$\tilde{X}_i \in \mathbb{R}^{n_i \times r_{\text{input}}}, \quad \tilde{Y}_i \in \mathbb{R}^{n_i \times r_{\text{output}}}.$$

The goal is to learn a global regression weight

$\beta_{\text{Fed}} \in \mathbb{R}^{r_{\text{input}} \times r_{\text{output}}}$ that minimizes the aggregated loss

$$\sum_{i=1}^C \|\tilde{X}_i \beta_{\text{Fed}} - \tilde{Y}_i\|_F^2,$$

without sharing raw embeddings.



Federated Linear Regression

Centralized SER (Linear Regression)

Low-dimensional SVD embeddings:

$$\tilde{X} \in \mathbb{R}^{n \times r_{\text{in}}} \longrightarrow Y \in \mathbb{R}^{n \times r_{\text{out}}}$$

Ridge Regression:

$$\min_{\beta} \|Y - \tilde{X}\beta\|_F^2 + \lambda \|\beta\|_F^2$$

Least Squares:

$$\hat{\beta} = (\tilde{X}^\top \tilde{X} + \lambda I)^{-1} \tilde{X}^\top Y$$

$$S = \tilde{X}^\top \tilde{X}, \quad B = \tilde{X}^\top Y$$

Federated SER (Linear Regression)

Client-local embeddings:

$$\tilde{X}_i \in \mathbb{R}^{n_i \times r_{\text{in}}}, \quad Y_i \in \mathbb{R}^{n_i \times r_{\text{out}}}$$

Gram-matrix aggregation:

$$S = \sum_{i=1}^C \tilde{X}_i^\top \tilde{X}_i, \quad B = \sum_{i=1}^C \tilde{X}_i^\top Y_i$$

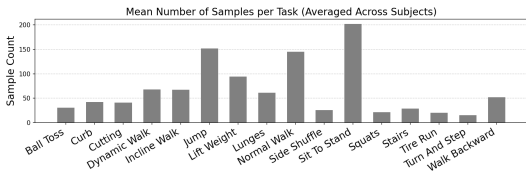
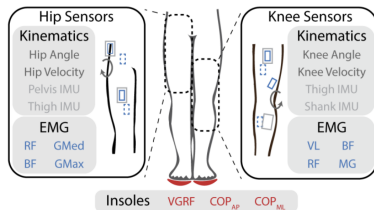
Server solves:

$$\hat{\beta}_{\text{Fed}} = (S + \lambda I)^{-1} B$$

No raw data shared; identical solution.

Dataset: Multi-Modal and Multi-Task Lower-Limb Biomechanics

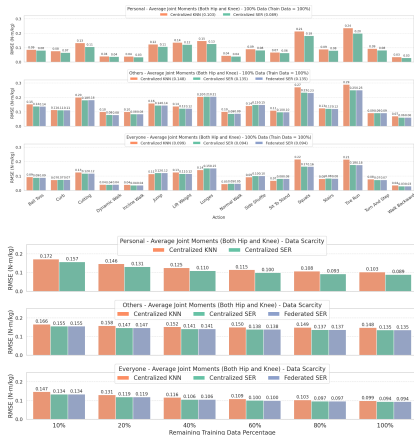
- 11 participants with $\sim 1k$ samples per participant.
- 16 shared activities; wide range of difficulty and intensity, cyclic & non-cyclic (e.g. walking, running, stairs, inclines, squats).
- Sensors** (wearable + reference)
 - IMUs (200 Hz): pelvis, bilateral thighs & shanks
 - EMG (2 kHz): 8 muscles/leg (bilateral)
 - Motion capture (200 Hz): reference kinematics
 - Force plates $\rightarrow$ virtual insoles: GRF & COP
- Inputs $\rightarrow$ Targets:** IMU, EMG, kinematics $\rightarrow$ joint moments (Hip & Knee)



Scherpereel, K. L., et al. "Improving biological joint moment estimation during real-world tasks with EMG and instrumented insoles." *IEEE TBME*, 2024.

Federated SER Achieves the Same Accuracy as Centralized SER

Federated SER matches centralized SER for all tasks using “Everyone” data, across all data scarcities.



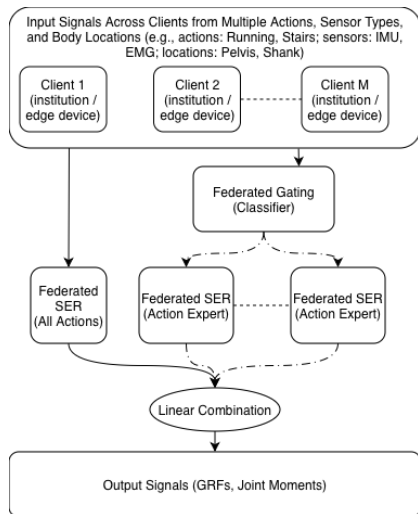
Extension to Multi-Task

Motivation

- One-model-per-action SER is more accurate than one-model-all-action SER.

Solution

- Add a lightweight **federated activity classifier** for gating — routing each input window to activity-specialized experts (optionally combined with a shared global model, useful for unseen actions).



Federated Classification — Problem Definition

A shared classifier over a global label space (random forest / ANN); a class-union-aware masked loss handles partial or disjoint labels across clients.

Federated setting: C clients. Client c holds a private local dataset

$$\mathcal{D}_c = \{(x_i^{(c)}, l_i^{(c)})\}_{i=1}^{n_c}, \quad x_i^{(c)} \in \mathbb{R}^d, \quad l_i^{(c)} \in \mathcal{L}_c.$$

Note: not all clients have all actions.

Shared representation via federated SVD:

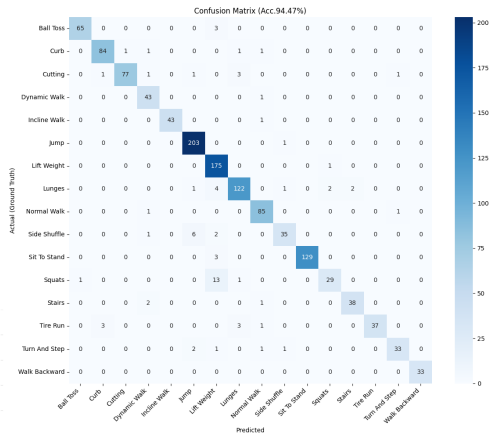
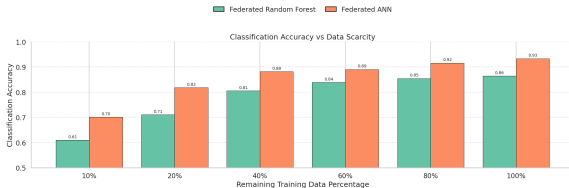
$$z = \phi(x) \triangleq V_{Fed}^\top x \in \mathbb{R}^k.$$

Downstream federated predictor options on embeddings z :

$$(i) \text{ Federated Random Forest: } \hat{l} = f_{RF}(z), \quad (ii) \text{ Federated ANN: } \hat{l} = f_{ANN}(z).$$

Federated Classification

- Accuracy remains high after **80%** of training data is removed.
- ANN is better than RF for all data scarcities.



Confusion matrix over 16 activities at 100% data availability.

Gating Strategies

- Let $p(a | x)$: the activity posterior produced by the gating classifier
- $f_a(\cdot)$: an activity-specific expert.
- $f_0(\cdot)$: A global model trained on all activities.

Gating strategies:

- **Hard gating (single expert selection):**

$$\hat{a} = \arg \max_a p(a | x), \quad \hat{y} = f_{\hat{a}}(x)$$

- **Linear-combination gating with global model:**

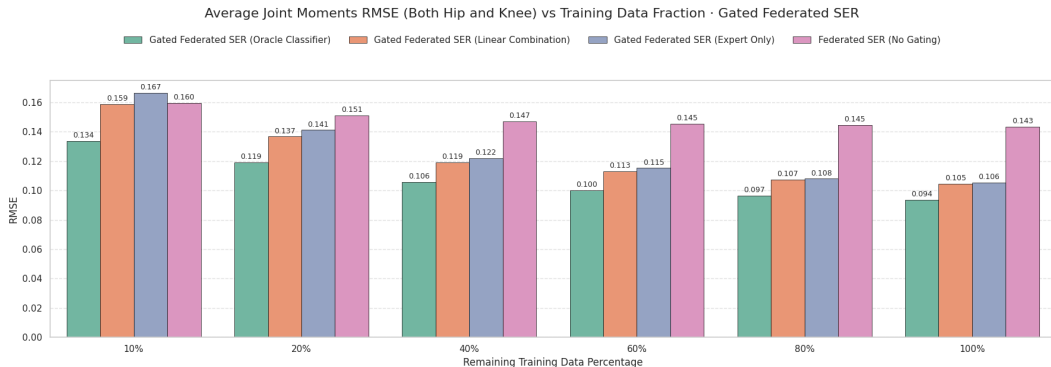
$$\hat{y} = \alpha(x) f_{\hat{a}}(x) + (1 - \alpha(x)) f_0(x), \quad \alpha(x) = \max_{a \in \mathcal{Y}} p(a | x) \in [0, 1],$$

The global model provides a shared baseline across tasks, while activity-specific experts refine predictions based on inferred context.

Deng, Y., Kamani, M. M., and Mahdavi, M. “Adaptive personalized federated learning.” arXiv:2003.13461 (2020).

Accuracies of Gated Federated SER vs. Data Scarcity

- Gating consistently improves performance relative to no gating.
- Linear combination consistently outperforms expert-only gating.

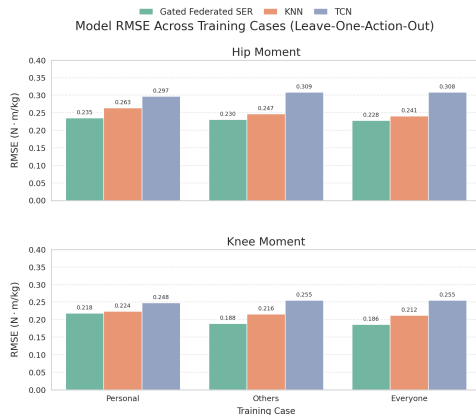


Seen and Unseen Tasks: Compare to State of the Art

Across almost all targets and regimes, SER achieves the lowest RMSE; TCN degrades, especially when actions are unseen during training.



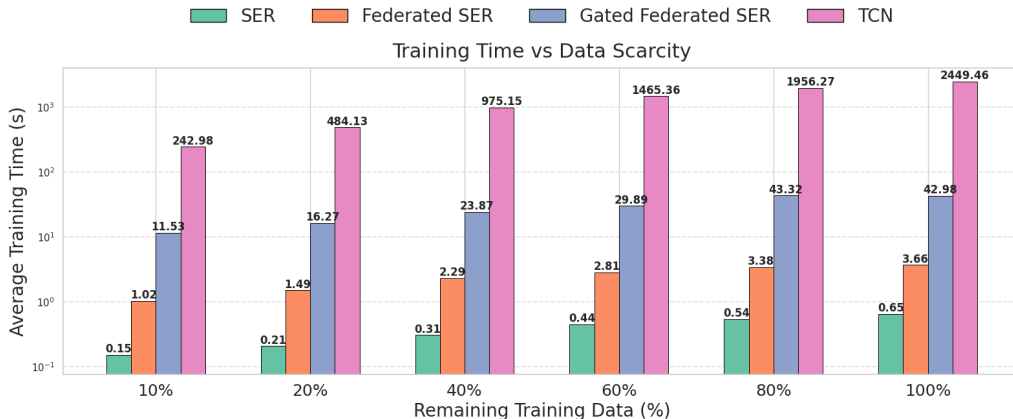
Seen tasks



Unseen tasks

Significantly Less Training Time

Training time is reported for a single finalized model, excluding hyperparameter tuning.



Summary of Contributions

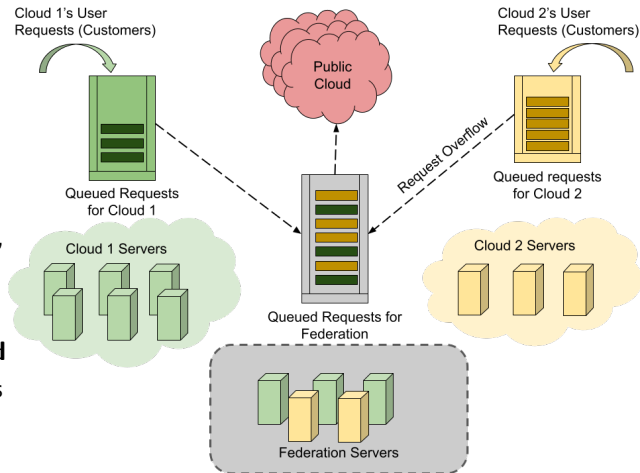
- **Efficient and lightweight:** FedSER retains the lightweight, low-latency nature of SER — enabling both federated training and inference on edge.
- **Privacy preserving:** clients send only second-order statistics to the server; supports add-ons such as secure aggregation and differential privacy.
- **Robust results:** robust to data scarcity and non-IID task distributions, staying accurate even when up to 90% of training tasks and samples are removed.
- Federated SER does not compromise accuracy compared to centralized SER.
- **Strong generalization** across tasks and subjects, especially in leave-one-action-out — consistently matching or outperforming TCN and KNN baselines.

Part III: Performance and Revenue Analysis of Hybrid Cloud Federations

Song, B., Paolieri, M., & Golubchik, L. (2022). *Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements*. In IEEE CLOUD 2022.

A Federation of Hybrid Clouds

- Researchers' workloads overflow from on-premises clouds to the costly public cloud.
- Hybrid cloud handles variable and peak loads only — it does not help over-provisioned clouds, nor under-provisioned ones with long-term growth.
- Motivates **resource & workload federation** with QoS guarantees that ensure mutual benefit (no participant worse off).

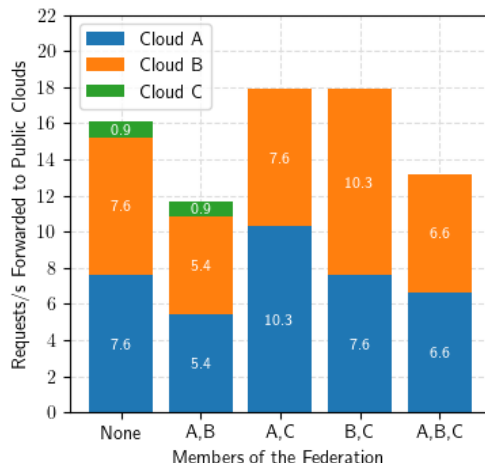


Motivating Example

Under the same arrival rate (100) and number of servers (100), each cloud has an individual latency requirement (QoS), in units of mean service time:

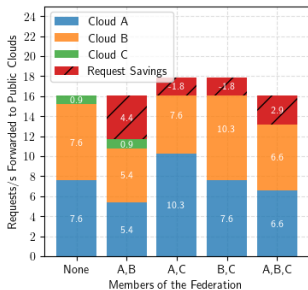
- **Cloud A:** 0 (no wait-time)
- **Cloud B:** 0 (no wait-time)
- **Cloud C:** 1 (wait up to 1 request)

{A, B} overflow only ~12 req/s to the public cloud — the best pairing; adding C helps less.

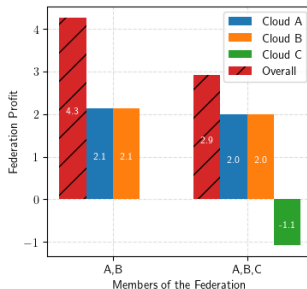


Main Questions

1. What is the optimal sharing strategy when forming a federation?
2. Any reason *not* to form a federation with certain clouds?
3. How to distribute federation revenue according to performance?



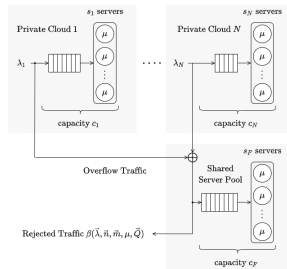
Public-cloud savings



Federation Profit Distribution

Performance Prediction

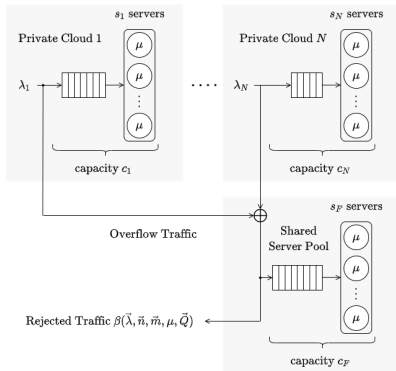
- Each cloud: **Poisson** arrivals, **exponential** service times; shares from nothing up to everything.
- Workloads queue locally first; if the local queue exceeds the latency requirement, they queue at the federation, then forward to the public cloud.
- The shared pool of servers and queueing space uses an **FCFS** policy.
- Best performance = least rejected (public-cloud) traffic.



For each cloud i :

- λ_i = arrival rate
- μ = service rate (globally homogeneous)
- n_i = total number of servers
- m_i = number of servers shared to the federation
- Q_i = latency requirement (QoS) in unit of mean service time

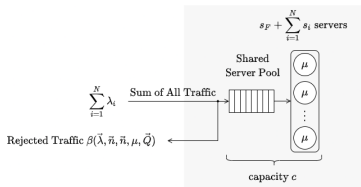
Optimal Sharing Strategy with Homogeneous QoS



Theorem

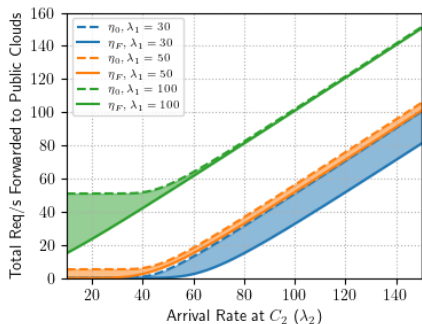
Under homogeneous QoS requirements, **full resource sharing** is the optimal strategy, yielding the minimum rate of rejected traffic.

- Sharing *any* amount yields non-decreasing benefit.

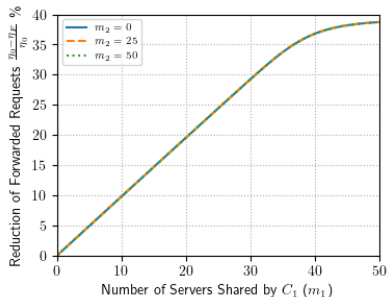


More Sharing, More Helpful

Best to group under-provisioned and over-provisioned clouds together when $Q_1 = Q_2$.



Given an over-provisioned (C_1) and an under-provisioned (C_2) cloud, sharing more of either reduces overall forwarding traffic.



Profit Sharing

- Cloud Profit: $CP = \text{revenue (const.)} - \text{operating cost (const.)} - \text{public-cloud cost}$; $\Delta CP = \text{cost before} - \text{cost after}$ federation.
- Federation Profit: $R(S) = P(\eta_0(S) - \eta_F(S)) = \sum_{i=1}^S \Delta CP_i$

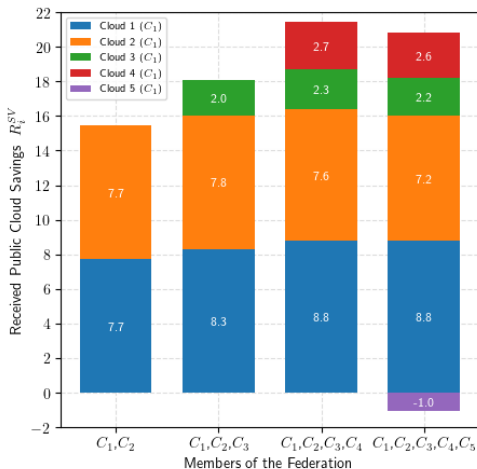
Reward policies

- $R_i^{SV}(S) := \sum_{S' \subseteq S \setminus \{i\}} \frac{|S'|! (|S| - |S'| - 1)!}{|S|!} (R(S' \cup \{i\}) - R(S'))$ (change in federation profit)
- $R_i^{SR}(S) := \frac{m_i}{\sum_{j=1}^K m_j} R(S)$ (shared resources)
- $R_i^{SIR}(S) := \frac{m_i(1 - \rho_i)}{\sum_{j=1}^K m_j(1 - \rho_j)} R(S)$ (shared idle resources)
- $R_i^{SIRW}(S) := \frac{1}{2} \left(R_i^{SIR}(S) + \frac{\lambda_i - \rho_i n_i \mu}{\sum_{j=1}^K \lambda_j - \rho_j n_j \mu} R(S) \right)$ (idle + traffic overflow)

Adding / Discouraging Members

- More clouds in the federation $\Rightarrow$ larger overall public-cloud savings.
- Shapley values compensate each cloud by its contribution relative to its peers.
- A cloud that *decreases* overall savings is discouraged (billed — the negative value), while others are rewarded to stay.

Cloud	Arrivals	Servers	QoS
1	10	30	0
2	50	30	0
3	30	30	0
4	50	50	0
5	50	50	1



Summary of Contributions

- Evaluate the effects of sharing strategies on the performance of federations.
- Prove that clouds with the same QoS can maximize federation performance and profit by **sharing everything**.
- Propose a profit-sharing mechanism that uses public-cloud savings to compute **Shapley values**, illustrating the ability to discourage “free riders”.

Thank You

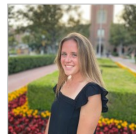
With thanks to my advisors, committee, and collaborators



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Questions?

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